

Question 1: Explain in your own words:

- **What is Data Science?**
 - **Why is it more than just statistics or programming?**
- Support your answer with one real world example.**

Answer:

Data Science is an interdisciplinary field that uses data, scientific methods, statistical techniques, and computer algorithms to extract meaningful insights and support decision-making. It involves collecting, cleaning, analyzing, and interpreting large amounts of structured and unstructured data to solve real-world problems.

In today's digital world, organizations generate huge volumes of data every day through websites, mobile applications, social media platforms, sensors, and business transactions. Data Science helps organizations convert this raw data into valuable information that can improve business performance, customer satisfaction, and operational efficiency.

Data Science is often misunderstood as only statistics or only programming. However, it is much more than either of these disciplines.

Statistics is an important part of Data Science because it helps in understanding data, identifying patterns, testing hypotheses, and making predictions. Statistical methods allow data scientists to draw conclusions from data and measure the reliability of their findings.

Programming is also a crucial component because modern datasets are often too large and complex to analyze manually. Programming languages such as Python and R help automate data collection, cleaning, visualization, and model development.

However, Data Science goes beyond statistics and programming because it also includes:

1. **Domain Knowledge** – Understanding the business or industry problem being solved.
2. **Data Engineering** – Collecting, storing, and preparing data for analysis.
3. **Machine Learning** – Building models that can learn patterns and make predictions automatically.
4. **Data Visualization** – Presenting findings through charts, graphs, and dashboards.
5. **Decision-Making** – Converting insights into practical business actions.

A Data Scientist must combine all these skills to create solutions that generate value from data.

Real-World Example

A popular example of Data Science is **Netflix's recommendation system**.

Netflix collects data about:

- Movies and shows watched by users
- Watch duration
- Ratings and reviews
- Search history
- Viewing preferences

Using Data Science techniques, Netflix analyzes this data and predicts what content a user is most likely to enjoy. Machine learning algorithms study user behavior patterns and generate personalized recommendations.

For example, if a user frequently watches action and science-fiction movies, Netflix will recommend similar content. This improves user satisfaction and encourages users to spend more time on the platform.

Thus, Data Science helps Netflix enhance customer experience, increase user engagement, and improve business growth.

Question 2: A retail company wants to:

- **Collect customer data**
- **Build a churn prediction model**
- **Deploy it in production**

Explain the role of: • Data Analyst • Data Scientist • Data Engineer • ML Engineer

At which stage does each contribute?

Answer

Building and deploying a customer churn prediction system is a collaborative effort involving multiple professionals. Each role contributes at different stages of the project lifecycle. Although their responsibilities may sometimes overlap, each professional has a specific focus that ensures the success of the entire system.

1. Data Engineer

The Data Engineer is responsible for designing and maintaining the data infrastructure required for the project. Since customer data often comes from multiple sources such as websites, mobile applications, CRM systems, purchase records, and customer support interactions, the Data Engineer ensures that all this data is collected, stored, and processed efficiently.

Their responsibilities include:

- Collecting customer data from different sources.
- Building data pipelines to automate data movement.
- Cleaning and transforming raw data.
- Managing databases and data warehouses.
- Ensuring data quality and reliability.
- Making data accessible for analysis and model development.

Without Data Engineers, the organization would struggle to handle large volumes of customer data effectively.

Stage of Contribution:
Data Collection and Data Preparation Stage

2. Data Analyst

Once the data is available, the Data Analyst examines it to identify trends, patterns, and business insights. Their goal is to understand customer behavior and provide valuable information to stakeholders.

In a churn prediction project, a Data Analyst may investigate:

- Which customers are leaving the company.
- Common characteristics of churned customers.
- Purchase behavior patterns.
- Customer satisfaction levels.
- Revenue impact of customer churn.

They use tools such as SQL, Excel, Power BI, and Tableau to create reports and dashboards that help management understand the problem.

Their findings often help Data Scientists decide which variables should be included in the predictive model.

Stage of Contribution:
Exploratory Data Analysis (EDA) and Business Analysis Stage

3. Data Scientist

The Data Scientist is responsible for developing the churn prediction model. They use statistical techniques and machine learning algorithms to predict whether a customer is likely to leave the company.

Their responsibilities include:

- Selecting relevant features from customer data.
- Performing feature engineering.
- Choosing suitable machine learning algorithms.
- Training and testing predictive models.
- Evaluating model performance.
- Improving model accuracy.

For example, a Data Scientist may use algorithms such as:

- Logistic Regression
- Decision Trees
- Random Forest
- XGBoost
- Neural Networks

The Data Scientist's primary goal is to create an accurate model that can predict customer churn before it happens.

Stage of Contribution:

Model Development and Model Evaluation Stage

4. ML Engineer (Machine Learning Engineer)

After the Data Scientist develops a successful model, the ML Engineer takes responsibility for deploying and maintaining it in a real-world environment.

Their responsibilities include:

- Deploying the model into production.
- Creating APIs for model access.
- Integrating the model with company applications.
- Monitoring model performance.
- Handling scalability issues.
- Updating and retraining models when needed.
- Ensuring reliability and security.

For example, when a customer logs into the company's website, the deployed model may instantly calculate the customer's churn risk and alert the customer retention team.

The ML Engineer ensures that the model continues to perform efficiently in real-world conditions.

Stage of Contribution:

Deployment, Monitoring, and Maintenance Stage

Workflow

Step 1: Data Collection

Data Engineer

- Collects customer data.
- Stores and processes it.

Step 2: Data Analysis

Data Analyst

- Studies customer behavior.
- Identifies churn patterns.
- Creates reports and dashboards.

Step 3: Model Building

Data Scientist

- Develops and trains churn prediction models.
- Evaluates and improves performance.

Step 4: Deployment

ML Engineer

- Deploys the model to production.
- Integrates it into business systems.

Step 5: Monitoring

ML Engineer

- Tracks model accuracy.
 - Updates models when required.
-

Question 3

A hospital wants to:

- **Predict patient readmission risk**
- **Automate report generation**
- **Deploy a chatbot for patient FAQs**

Map each problem to:

- **Traditional Data Science**
- **Machine Learning**
- **GenAI**
- **Agentic AI**

Explain your reasoning.

Answer

Healthcare organizations generate large amounts of data every day, including patient records, medical reports, treatment histories, laboratory results, and feedback. Modern Artificial Intelligence technologies help hospitals improve efficiency, reduce costs, and provide better

patient care. Different healthcare problems require different AI approaches depending on the nature of the task.

In this scenario, the hospital wants to predict patient readmission risk, automate report generation, and deploy a chatbot for patient FAQs. These tasks can be mapped to Traditional Data Science, Machine Learning, Generative AI (GenAI), and Agentic AI as follows.

1. Predict Patient Readmission Risk → Machine Learning

Predicting whether a patient is likely to be readmitted to the hospital after discharge is a **Machine Learning** problem.

Machine Learning involves training algorithms on historical data so they can identify patterns and make predictions about future events.

The hospital can use data such as:

- Patient age
- Medical history
- Existing diseases
- Treatment records
- Length of hospital stay
- Previous admissions
- Medication details

Using this data, a machine learning model learns patterns associated with patient readmissions.

For example, if patients with certain medical conditions and treatment histories frequently return within 30 days, the model can identify similar high-risk patients in the future.

Common algorithms used include:

- Logistic Regression
- Decision Trees
- Random Forest
- XGBoost
- Neural Networks

Why Machine Learning?

Because the goal is to **predict a future outcome based on historical data**, which is the primary purpose of machine learning.

Benefits

- Early identification of high-risk patients.
- Reduced healthcare costs.
- Improved patient care.
- Better resource allocation.

2. Automate Report Generation → Generative AI (GenAI)

Generating patient reports automatically is a **Generative AI (GenAI)** application.

Generative AI specializes in creating new content such as:

- Text
- Images
- Audio
- Code

In a hospital, doctors and nurses generate large amounts of documentation daily. Writing reports manually can be time-consuming.

A GenAI system can:

- Read patient records.
- Analyze test results.
- Summarize treatment history.
- Generate discharge summaries.
- Create medical reports in natural language.

For example:

Input Data:

- Blood Pressure: Normal
- Blood Sugar: Elevated
- Diagnosis: Type 2 Diabetes

Generated Report:

"The patient was diagnosed with Type 2 Diabetes. Blood pressure levels remained within the normal range, while blood glucose levels were elevated. Appropriate medication and lifestyle recommendations have been provided."

Why GenAI?

Because the system is **creating human-like text content automatically**, which is the primary function of Generative AI.

Benefits

- Saves doctors' time.
- Reduces documentation workload.
- Improves report consistency.
- Faster patient processing.

3. Chatbot for Patient FAQs → Agentic AI

A patient FAQ chatbot is best classified as **Agentic AI**.

Agentic AI refers to AI systems that can:

- Understand goals.
- Make decisions.
- Take actions.
- Interact autonomously with users.

A hospital chatbot may:

- Answer patient questions.
- Schedule appointments.
- Check appointment status.
- Provide medication reminders.
- Guide patients to relevant departments.
- Escalate complex cases to human staff.

For example:

Patient: *"When is my next appointment?"*

Agentic AI Chatbot:

- Retrieves patient information.
- Checks appointment records.
- Responds with the correct date and time.
- May even reschedule the appointment if requested.

Unlike simple rule-based chatbots, Agentic AI can perform tasks and make decisions on behalf of the user.

Why Agentic AI?

Because the system not only generates responses but can also **perform actions and make decisions autonomously**.

Benefits

- 24/7 patient support.
- Reduced workload for hospital staff.
- Faster response times.
- Improved patient experience.

4. Traditional Data Science in the Hospital

Traditional Data Science focuses on analyzing historical data to discover trends, patterns, and insights that support decision-making.

Examples include:

- Identifying common diseases.
- Analyzing patient demographics.
- Tracking hospital occupancy rates.
- Studying treatment effectiveness.
- Measuring patient satisfaction.

For example, hospital administrators may analyze five years of patient data to determine which departments experience the highest patient load during specific seasons.

Why Traditional Data Science?

Because the goal is to **understand and interpret data rather than make predictions or generate content**.

Benefits

- Better planning and management.
- Data-driven decisions.
- Improved operational efficiency.
- Enhanced healthcare quality.

Question 4

An e-commerce company wants an AI system that:

- Monitors inventory
- Predicts demand
- Automatically places supplier orders
- Negotiates price via email

Explain:

1. Why this is an example of Agentic AI.
2. What risks might arise?
3. What human oversight would still be required?

Answer

Artificial Intelligence has evolved from systems that only analyze data to systems that can make decisions and perform actions independently. Such systems are known as **Agentic AI**. Unlike traditional AI models that only provide recommendations, Agentic AI can observe its environment, make decisions, take actions, and continuously adapt to changing situations with minimal human intervention.

In this scenario, an e-commerce company wants an AI system that can monitor inventory, predict future demand, automatically place supplier orders, and negotiate prices via email. This system demonstrates several characteristics of Agentic AI.

1. Why is this an Example of Agentic AI?

Agentic AI refers to intelligent systems that can perform tasks autonomously by making decisions and taking actions to achieve specific goals.

The primary goal of the e-commerce company is to ensure that products are always available while minimizing inventory costs. The AI system achieves this goal through multiple connected activities.

Inventory Monitoring

The AI continuously tracks:

- Current stock levels
- Product availability
- Sales trends
- Warehouse data

Instead of waiting for a human employee to check inventory, the AI performs this task automatically.

Demand Prediction

The AI analyzes:

- Historical sales data
- Seasonal trends
- Customer purchasing behavior
- Market demand

Using machine learning models, it predicts future product demand.

For example, if the system notices that sales of air conditioners increase during summer, it can predict higher future demand and prepare inventory accordingly.

Automatic Supplier Orders

After predicting demand, the AI can decide:

- Which products need restocking
- How many units should be ordered
- When the order should be placed

The system does not simply suggest an order; it takes action by generating and submitting purchase orders automatically.

Price Negotiation

The AI can communicate directly with suppliers through email.

It may:

- Compare supplier quotations
- Request discounts
- Negotiate delivery schedules
- Finalize purchase terms

This ability to interact, reason, and make decisions independently makes the system an example of Agentic AI.

Why It Is Agentic AI

The system:

- Observes the environment.
- Analyzes data.
- Makes decisions.
- Takes actions.
- Communicates with external entities.
- Works toward achieving business goals autonomously.

These characteristics distinguish Agentic AI from traditional AI systems.

2. What Risks Might Arise?

Although Agentic AI offers significant benefits, it also introduces several risks that organizations must carefully manage.

Incorrect Demand Predictions

The AI may incorrectly estimate future demand due to:

- Sudden market changes
- Economic conditions
- New competitors
- Unexpected customer behavior

This could result in:

- Overstocking
- Understocking
- Increased costs

Financial Losses

If the AI orders excessive inventory, the company may face:

- Higher storage costs
- Unsold inventory
- Reduced profits

Similarly, ordering too little inventory may lead to lost sales opportunities.

Poor Supplier Negotiations

The AI may negotiate unfavorable terms such as:

- Higher prices
- Longer delivery times
- Poor contract conditions

This could negatively affect business operations.

Data Quality Issues

AI decisions are only as good as the data provided.

Problems may occur if:

- Inventory data is inaccurate
- Sales records are incomplete
- Supplier information is outdated

Poor-quality data can lead to poor decisions.

Security Risks

If hackers gain access to the AI system, they may:

- Manipulate orders
- Access confidential information
- Cause financial damage

Cybersecurity becomes critical when AI systems are connected to suppliers and business operations.

Ethical and Compliance Risks

The AI may unintentionally:

- Violate procurement policies
- Make biased decisions
- Use unauthorized supplier information

Such issues could create legal and reputational problems for the company.

3. What Human Oversight Would Still Be Required?

Even though Agentic AI can operate autonomously, human supervision remains essential.

Approval of High-Value Purchases

Large purchase orders should require human approval before execution.

Managers can verify:

- Quantity
- Cost

- Supplier selection

before the order is finalized.

Monitoring AI Decisions

Business teams should regularly review:

- Demand forecasts
- Inventory decisions
- Supplier negotiations

to ensure the AI is performing correctly.

Exception Handling

Some situations require human judgment, such as:

- Supply chain disruptions
- Natural disasters
- New product launches
- Unexpected market trends

Humans can evaluate complex situations that AI may not fully understand.

Performance Audits

Organizations should periodically assess:

- Forecast accuracy
- Cost savings
- Supplier satisfaction
- Business outcomes

This helps identify and correct issues in the system.

Compliance and Ethical Review

Humans should ensure that AI decisions comply with:

- Company policies
- Industry regulations
- Ethical standards

This prevents legal and reputational risks.

Final Accountability

Regardless of how autonomous the AI becomes, responsibility for business decisions remains with human managers and organizational leadership.

Advantages of Agentic AI in E-Commerce

- Faster decision-making.
 - Reduced manual workload.
 - Improved inventory management.
 - Better demand forecasting.
 - Cost savings.
 - Increased operational efficiency.
 - Improved supplier communication.
-

Question 5: Given:

```
price = 499
discount = 0.15
final_price = price - price * discount
```

1. What is the data type of each variable?
2. Modify the code so that `final_price` is printed with 2 decimal places.
3. Convert `price` into a string and concatenate it into:
"The final price is ₹__"

Answer

Python is a high-level programming language that supports different data types for storing and manipulating information. Every variable in Python stores a value of a specific data type, and understanding these data types is important for performing calculations and data processing correctly.

In the given program, three variables are used: `price`, `discount`, and `final_price`.

1. Data Type of Each Variable

Variable: price

price = 499

The value 499 is a whole number without a decimal point.

Therefore, its data type is:

int

(Integer)

Integers are used to store whole numbers such as:

10

25

100

499

Variable: discount

discount = 0.15

The value contains a decimal point.

Therefore, its data type is:

float

(Floating Point Number)

Floats are used to store decimal values such as:

3.14

0.15

99.99

12.5

Variable: `final_price`

```
final_price = price - price * discount
```

Calculation:

$$499 \times 0.15 = 74.85$$
$$499 - 74.85 = 424.15$$

Since the result contains decimal values, Python stores it as:

float

2. Print `final_price` with 2 Decimal Places

In business and financial applications, prices are usually displayed with two decimal places.

Python provides **formatted strings (f-strings)** to control the display format.

Code

```
price = 499

discount = 0.15

final_price = price - price * discount

print(f"Final Price: ₹{final_price:.2f}")
```

Output

Final Price: ₹424.15

Explanation

The formatting expression:

`:.2f`

means:

- `:` → Start formatting
- `.2` → Display 2 digits after decimal
- `f` → Floating-point format

Example:

```
num = 25  
  
print(f'{num:.2f}')
```

Output:

25.00

3. Convert **price** into a String and Concatenate It

Sometimes we need to combine numbers with text.

Since Python cannot directly join integers and strings, we must first convert the integer into a string using the `str()` function.

Code

```
price = 499  
  
message = "The final price is ₹" + str(price)  
  
print(message)
```

Output

The final price is ₹499

Using **final_price**

A more practical example is:

```
price = 499  
  
discount = 0.15
```

```
final_price = price - price * discount
```

```
message = "The final price is ₹" + str(final_price)
```

```
print(message)
```

Output

The final price is ₹424.15

Question 6: A company gives bonus based on performance score:

- **Score ≥ 90 : 20% bonus**
- **Score ≥ 75 : 10% bonus**
- **Else : No bonus**

Write Python logic using:

- **Comparison operators**
- **Logical operators**
- **Conditional statements**

Answer:

In a company, employees may receive bonuses based on their performance score. Python provides comparison operators, logical operators, and conditional statements to handle such situations. Comparison operators are used to compare values. For example, `>=` checks whether a score is greater than or equal to a particular value. Logical operators such as `and`, `or`, and `not` are used when multiple conditions need to be checked together. Conditional statements like `if`, `elif`, and `else` help the program make decisions based on the result of these conditions.

For the given bonus policy, if an employee scores 90 or more, they receive a 20% bonus. If the score is between 75 and 89, they receive a 10% bonus. Otherwise, no bonus is given.

Code:

```
score = 85

if score >= 90:
    print("20% Bonus")
elif score >= 75:
    print("10% Bonus")
else:
    print("No Bonus")
```


In the above code, the program first checks whether the score is 90 or above. If the condition is true, it prints 20% bonus. If not, it checks whether the score is 75 or above. If that condition is true, it prints 10% bonus. If neither condition is true, the program prints "No Bonus". This type of logic is commonly used in payroll systems, employee management software, and performance evaluation systems.

Question 7: You are given a customer review:

```
review = "The product quality is Excellent but delivery was Late"
```

Write Python code to:

1. Convert to lowercase
2. Count number of words
3. Check if the word "excellent" exists
4. Replace "Late" with "delayed"

Answer :

In Python, strings are used to store and manipulate text data. Customer reviews are often analyzed by businesses to understand customer satisfaction and identify areas for improvement. Python provides several built-in string functions that make text processing easy and efficient.

In this example, a customer review is given. We need to perform different operations such as converting the text to lowercase, counting the number of words, checking whether a specific word exists, and replacing a word with another word.

Code:

```
review = "The product quality is Excellent but delivery was Late"
```

```
# 1. Converting to lowercase
```

```
lower_review = review.lower()
```

```
print("Lowercase Review:", lower_review)
```

```
# 2. Count number of words
```

```
word_count = len(review.split())
```

```
print("Number of Words:", word_count)
```

```
# 3. Check if 'excellent' exists
```

```
print("excellent" in lower_review)
```

```
# 4. Replace 'Late' with 'delayed'
```

```
updated_review = review.replace("Late", "delayed")
```

```
print(updated_review)
```

Output

Lowercase Review: the product quality is excellent but delivery was late

Number of Words: 9

True

The product quality is Excellent but delivery was delayed

In the Code :

First, the `lower()` function is used to convert all characters in the review to lowercase. This is useful because it makes text analysis easier and avoids issues caused by different letter cases.

Next, the `split()` function divides the sentence into individual words, and the `len()` function counts the total number of words present in the review. In this case, there are 9 words.

To check whether the word "excellent" exists in the review, the `in` operator is used. Since the review was converted to lowercase first, the check becomes case-insensitive and returns True.

Finally, the `replace()` function is used to replace the word "Late" with "delayed". This creates a new string where the specified word is updated without changing the original review.

Question 8: Given sales data:

```
sales = [12000, 15000, 11000, 18000, 17000]
```

1. Calculate total revenue.
2. Find average revenue.
3. Identify highest sales month.
4. Append new month sale = 20000.
5. Sort the list in descending order.

Answer :

Sales analysis is an important part of every business because it helps companies understand their performance and make better decisions. In Python, lists can be used to store sales data, and various built-in functions can be used to perform calculations and analysis.

In the given example, sales data for five months is stored in a list. We need to calculate the total revenue, average revenue, highest sale, add a new sales value, and sort the data in descending order.

Code:

```
sales = [12000, 15000, 11000, 18000, 17000]
```

```
# 1. Calculate total revenue
```

```
total_revenue = sum(sales)
```

```
# 2. Calculate average revenue
```

```
average_revenue = total_revenue / len(sales)
```

```
# 3. Find highest sales month
```

```
highest_sale = max(sales)

# 4. Append new month sale
sales.append(20000)

# 5. Sort in descending order
sales.sort(reverse=True)

print("Total Revenue:", total_revenue)
print("Average Revenue:", average_revenue)
print("Highest Sale:", highest_sale)
print("Updated Sales List:", sales)
```

Results

- Total Revenue = ₹73,000
- Average Revenue = ₹14,600
- Highest Sale = ₹18,000
- New Sale Added = ₹20,000
- Sorted Sales List = [20000, 18000, 17000, 15000, 12000, 11000]

Question 9: You are building a simple sales analysis system.

Given:

```
products = ["Laptop", "Phone", "Tablet"]
prices = [50000, 20000, 15000]
quantities = [5, 10, 7]
```

Tasks:

1. Compute total revenue for each product.
2. Identify which product generated maximum revenue.
3. Store revenues in a new list.
4. Print a formatted summary like:

In business and sales analysis, revenue is one of the most important metrics used to measure performance. Revenue is calculated by multiplying the price of a product by the quantity sold. Python makes it easy to perform such calculations using lists and loops.

In this example, we are given three lists containing product names, prices, and quantities sold. The objective is to calculate the revenue generated by each product, identify the product with the highest revenue, and store all revenue values in a separate list.

Code

```
products = ["Laptop", "Phone", "Tablet"]
```

```
prices = [50000, 20000, 15000]
```

```
quantities = [5, 10, 7]
revenues = []
for i in range(len(products)):
    revenue = prices[i] * quantities[i]
    revenues.append(revenue)
max_revenue = max(revenues)
max_product = products[revenues.index(max_revenue)]
print("Product Summary")
print("-----")
for i in range(len(products)):
    print(products[i], "- Revenue: ₹", revenues[i])
print("\nHighest Revenue Product:", max_product)
```

Question 10: A startup collects customer feedback as text and numerical ratings.

Explain step-by-step:

1. How raw text becomes structured data.
2. Where Python (strings & lists) is used.
3. Where Data Science models are used.
4. Where GenAI can enhance the system.
5. What ethical concerns must be addressed.

Answer:

Customer feedback is very valuable for a startup because it helps the company understand customer satisfaction, identify problems, and improve products or services. However, feedback is often collected in the form of text reviews and numerical ratings, which need to be processed before useful insights can be obtained. Data Science, Python, and Artificial Intelligence play an important role in this process.

1. How Raw Text Becomes Structured Data

Customer feedback is usually received as unstructured text such as:

"Great product and fast delivery"

This text cannot be directly analyzed by a machine. Therefore, it must first be converted into structured data.

The process includes:

- Collecting customer reviews.
- Removing unnecessary symbols and punctuation.
- Converting text into lowercase.
- Splitting sentences into individual words.
- Extracting useful keywords.
- Converting text into numerical values that computers can understand.

For example:

Review: "Great product and fast delivery"

Structured Data:

Positive Words = 2

Negative Words = 0

Rating = 5

After this process, the data can be stored in tables or databases for further analysis.

2. Where Python (Strings & Lists) is Used

Python is widely used for handling customer feedback data.

Strings are used to store and process text reviews.

Example:

```
review = "Great product and fast delivery"  
print(review.lower())
```

Lists are used to store multiple reviews or ratings.

Example:

```
reviews = [  
    "Great product",  
    "Good service",  
    "Fast delivery"  
]  
ratings = [5, 4, 5]
```

Python functions help in:

- Cleaning text
- Counting words
- Finding keywords
- Storing and organizing feedback data

Because of its simplicity, Python is one of the most popular languages in Data Science.

3. Where Data Science Models are Used

Once the data becomes structured, Data Science models can be applied to analyze customer opinions and identify patterns.

Some common applications include:

- Sentiment Analysis
- Customer Satisfaction Prediction
- Trend Analysis
- Product Performance Evaluation

For example, a sentiment analysis model can determine whether a review is:

- Positive
- Negative
- Neutral

Example:

Review: "Excellent product"

Prediction: Positive

The company can then understand how customers feel about its products and services.

4. Where GenAI Can Enhance the System

Generative AI (GenAI) can make the feedback system more powerful by generating human-like responses and summaries.

GenAI can:

- Summarize thousands of customer reviews.
- Generate feedback reports automatically.
- Suggest responses to customer complaints.
- Create improvement recommendations.
- Answer customer queries through chatbots.

For example, if a company receives 1,000 reviews, GenAI can generate a summary such as:

"Most customers are satisfied with product quality, but many have reported delayed deliveries."

This saves time and helps management make faster decisions.

5. Ethical Concerns That Must Be Addressed

While analyzing customer feedback, several ethical issues must be considered.

Privacy Protection

- Customer personal information should remain secure.
- Sensitive data should not be shared without permission.

Data Security

- Feedback data should be protected from hackers and unauthorized access.

Bias in AI Models

- AI systems should treat all customers fairly.

- Models should not favor or discriminate against any group.

Transparency

- Customers should know how their feedback is being used.

Responsible Use of Data

- Feedback should only be used for business improvement and customer service purposes.

Addressing these concerns helps build customer trust and ensures ethical use of Artificial Intelligence.
